

Using RGB-D Sensors for the Detection of Abandoned Luggage

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Abstract

In this paper we present a novel approach for the detection of abandoned luggage in an intelligent video surveillance system. Abandoned luggage, especially in critical infrastructures such as airports and train stations, are always dealt with as a priority security threat, and therefore an early detection is crucial for a swift reaction and evacuation initiated by the security personnel. The approach that will be presented in this paper is capable of functioning in dynamic environments and can handle luggage pieces of various sizes and colours. The novelty of this approach is in combining the data of 2 sensors, the **RGB sensor** and the **depth sensor**, to achieve a robust foreground segmentation. The novelty also lies in the use of the **RGB sensor** to extract the features of the suspected abandoned piece of luggage and verify them in the following frames. The tracking algorithm of the **depth sensor** is used to detect users and therefore eliminate persons from the list of suspected objects and reduce the amount of objects to be observed and analysed. Finally, we have evaluated the algorithm in an installation inside a fully functional tram within the frameworks of a German nationally funded project called InREAKT in three iterations.

1 Introduction

Occurrences such as the failed bomb detonation attempt in the central railway station of Dresden in 2003 [1] and on 2 regional trains for the Deutsche Bahn in 2016 [2], when suitcases armed with explosive material were abandoned with the intent to cause harm; show the necessity of an automated system to detect such threats. Surveillance staff are often overwhelmed by amount of data generated through the numerous CCTV cameras. This overwhelming situation makes it difficult to spot and identify an abandoned piece of luggage without any assisting system. For this reason, this paper will present the methodology and results of a detection module within a video surveillance system, which aims to assist the security personnel in detection and localising potential abandoned pieces of luggage and thus shortening the time leading to an appropriate reaction such as an evacuation. This module makes use of the state of the art depth sensor technology in combination with a RGB sensor and a state of the art feature extraction algorithm to robustly detect unattended luggage in an urban environment.

This module was implemented as part of an automated safety system proposed by the nationally funded project InREAKT [3].

2 Related work

The subject of finding and identifying unattended luggage with the help of cameras was often a matter of research in the past. The common approach is to identify an abandoned object within a scene and measure the time in which the object stays stationary. The aim is raising an alarm or notifying security personnel after a certain period of time has elapsed.

Detecting and tracking a piece of luggage is not a trivial task within an environment with fast lighting changes, dynamic items in the scene, shadows, reflections, and moving background objects. It can be therefore very challenging to implement a system, which is robust and can operate without interruption with a reasonable computational load. Lin et al. use finite-state-machine model to extract the foregrounds of a scene and even track the owner of abandoned luggage [4]. Liao et al. place emphasis on foreground-mask sampling for their abandoned luggage detection [5], and Schreiber et al. feature the usage of depth-cameras [6]. The unattended luggage detection feature of the InREAKT project is a combination of a multi-sensor approach which places the emphasis on the detected abandoned objects without analysing any personal data of any involved parties in the scene for data privacy reasons.

3 Methodology

The methodology of automatically acknowledging and verifying an object as an abandoned luggage can be divided into two steps (see Figure 1). The first step is object segmentation: the region of interest (ROI) which contains an object that ought to be observed (suspicious objects) has to be identified in each new frame over a specific period of time. Thereafter, the location and the features for each suspected object in a frame are saved for later verification in the second step.

The second step is feature detection and matching: a continuous analysis is applied, in order to detect whether the suspicious objects detected in step one were moved, had contact with humans, or are still in the field of vision over time. Both steps make use of the RGB- and depth-data of the Xtion PRO LIVE sensor (Xtion PRO LIVE is equipped with an infra-red sensor, adaptive depth detection technology, and a RGB sensor), to increase the robustness of the whole system and to avoid false-

positive outputs. If an object was detected as suspicious, an alarm will guide the attention of the security personnel to the specific scene, where the potential piece of unattended luggage is to be found. To explain the used methods and algorithms, the two steps of this feature are described in detail.

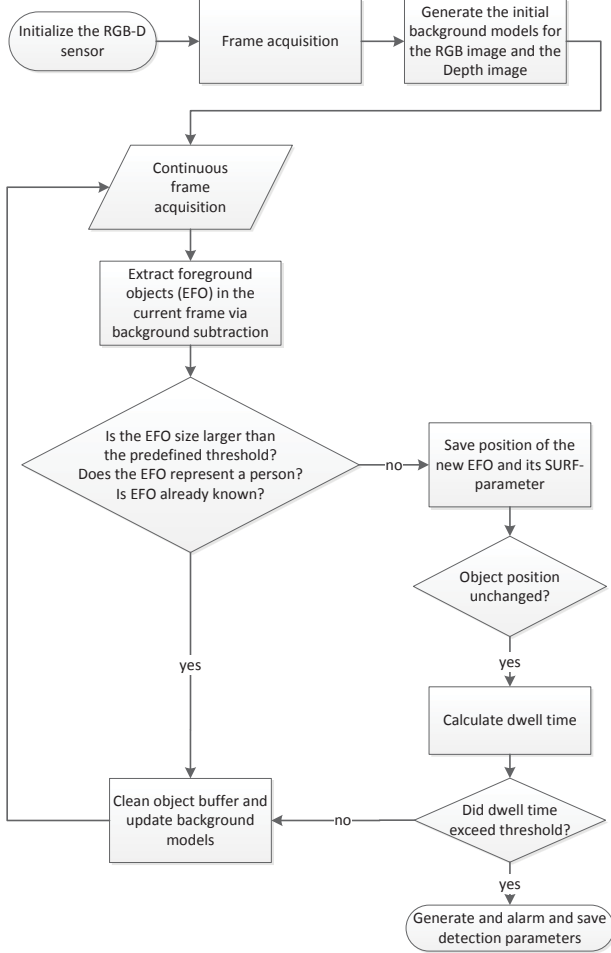


Figure 1: The work flow of the abandoned luggage detection module.

3.1 Object Segmentation

To identify objects within a frame, a background-subtraction algorithm was implemented. The correct execution of the background-subtraction is the most crucial part of the system. To ensure the correct creation of the background models, the approaches from other scientific projects [7] were used and adjusted to our needs. The initial background model is a result from the usage of a median value for each pixel, over a series of 50 frames (1).

$$B_{median}(x, y, t) = median\{V(x, y, t - i)\} \quad (1)$$

where N is the number of successive frames taken for averaging (50 frames for our system). This averaging refers to the averaging of corresponding pixels in these successive frames. N would also depend on the video speed (frames/sec) and the

amount of movement in the frame sequence. This way, people who move through the scene during the creation of the initial background, won't cause much disturbance. To avoid the decay of a background model through changes in the scenes background, it has to be adaptable to these changes.

To counter the effects that a motionless object gets integrated in the background model over time while the ROI is extracted, three background models are used, which all adapt in a different way. The first background model is a continuation of the initial median-background model (B_{median}). With each frame update, every new pixel value is stored in a circular buffer, which is then checked for the median value. This is a computational expensive but efficient way to update a background model [8]. The other two background models (bg) use a running Gaussian average algorithm (2) [8] with different adaptation rates α for foreground and background areas in the current frame.

$$bg_{i+1}(x, y) = \alpha \cdot frame_t(x, y) + (1 - \alpha) \cdot bg_i(x, y) \quad (2)$$

The adaptation rate determines how fast the pixel value of a background model changes and is integrated into the background model. The secondary background model (bg_{sec}) stores its information for a long time, since it has a very low adaptation rate α and only updates those pixels, which are considered background in the **current frame**. It prevents the blending of motionless objects into the background model.

The primary background model adapts faster than the secondary, but slower than the median background model. All areas of the primary background model adapt in dependency of the current frame, but those areas considered foreground adapt slower, i.e. a lower adaptation rate α is used. The primary background (bg_{prim}) model is the model which best represents the current situation. It will be overwritten with the data of one of the other background models, in case they are closer to reality. To check, which of the background models is best suited for the background subtraction, the histogram of each model is compared to the histogram of the current frame. The compareHist-function within the OpenCV library has been used for this purpose and the correlation comparison value is evaluated (CV_COMP_CORREL). This coordinated update mechanism ensures that the background model used resembles the real background as precise as possible to insure a better segmentation results.

The subtraction of the primary background model from the current frame leaves a difference image which is in most cases very noisy. To cancel out the effects of camera noise This method suppresses incorrect classification of pixels with the threshold-with-hysteresis which is more accurate than morphological methods [9]. The result of the hysteresis method is a mask, which contains all areas of the current frame, which are considered as foreground, since foreground areas in the current frame might represent an abandoned piece of luggage (suspicious object).

To increase the robustness of the whole system, the procedure of background model generation and foreground extraction described above, will be applied to the raw data of both the

RGB- and the depth-sensor. The results are two masks, which will be then merged to get hold of all areas of the current frame, which represent the foreground median (3).

$$fg_{merged} = fg_{rgb} \vee fg_{depth} \quad (3)$$

That way, both sensors complement each other and increase the chance of detecting all foreground objects within the current frame. Furthermore, using both sensors can prevent wrong foreground classifications caused by sudden lighting changes, since the depth sensor is much resilient to artefacts caused by the variation of elimination. The sensitivity of the RGB sensor and resilience of the depth sensor enables the detection of sudden light changes. This detection is achieved by comparing the sum of the areas segmented using the RGB-data with the sum segmented using the depth-data. If the difference is more than 20 % then the background model is updated and the current RGB-background estimation is discarded (see Figure 2).

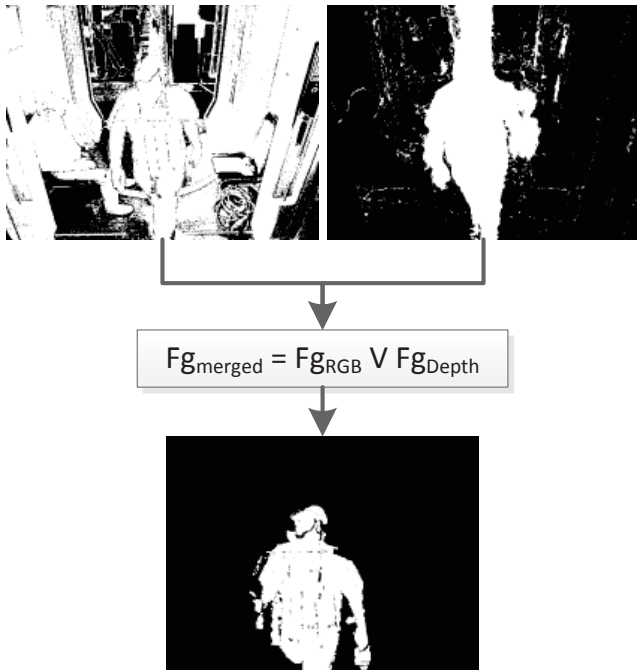


Figure 2: Merging the foreground images to produces a clean and robust foreground extraction.

After the creation of such a merged mask with results of the background subtraction of both sensors, the foreground areas are checked for their size using `minEnclosingCircle` (OpenCV) and if they depict a human using the correlation with the `userMap` (OpenNI). Every area, which isn't large enough, or is part of a human is not considered as possible threat and is ignored in the further process of the program. After that, the foreground mask is altered by discarding all foreground elements which are too small to be a potential piece of abandoned luggage. This is necessary in order to ensure that the following adaptation process of the background models works properly.

A wrong classification of fore- and background areas of the current frame could negatively influence the development of

the background models. All contours large enough to be a potential piece of luggage are cropped out of the current RGB-frame. For this task a rectangular region with the height and length of the referred to object is used. Further more, the centroid of the rectangular region and its distance to the camera are saved in a temporary buffer, which is cleared after every program iteration.

3.2 Feature detection and matching

In this step of the algorithm, the data from the object detection step is evaluated and viewed over a period of time. For that, the RGB-data from the *temporary buffer* containing the suspected objects are verified in the current frame using the SURF algorithm (Speeded Up Robust Features), which is a scale- and rotation-invariant detector and descriptor [10]. The descriptor extracts a feature vector by describing the distribution of the intensity content within the interest point neighbourhood similar to the gradient information extracted by the older SIFT algorithm (Scale-invariant Feature Transform) [11].

The best features are then stored with the location data in the vector of current suspicious objects. To avoid, that an object appears more than once in the database and thus leading to more computational effort, the SURF features of each segmented object in the *temporary buffer* are compared to those objects in the aforementioned vector using SURF detection. If no accordance is found, a new entry is generated, where all feature point parameters, RGB-data of the object and its position data from the temporary storage are transferred to the *temporary buffer*. After that, the SURF feature points of each entry of the database are used to check whether or not an object is still in the image ROI of the current frame, in which it was originally found. Furthermore, the position and the human interacting with the detected object are observed. If the centroid of a matched object hasn't moved, a main dwelling counter is incremented. If the counter reaches a defined threshold, the object is considered abandoned. This threshold time depends on the experience of the operator of the surveillance system.

In the event of another object, or person blocking the view of the observed object, a hidden dwelling counter is incremented, whose value is used to update the main counter, as soon as the object is visible and verified again. Blocking the view to the observed object can be easily acknowledged using the depth data, as the stored depth value of the centroid of the object differs from the depth value of that position in the current frame. A smaller depth value indicates that there is a distinct object or a person between the object and the camera. If an object was removed and is no longer in the image section of the scene, the main counter decreases twice as fast as it increases. If it falls below a certain threshold, the corresponding object is deleted from the database.

Its critical for this algorithm to filter out the number of suspicious objects accurately and thus keeping it low, since the feature point detection with the SURF algorithm is computationally intensive.

4 Result

The algorithm was tested in an operational environment inside a fully functional tram within the framework of the nationally funded project InREAKT. The evaluation was done in three iterations. After each iteration the system has been optimized and improved. An example of an installation of the RGB-D sensor can be seen in Figure 3 to monitor an area of the wagon near the doors.

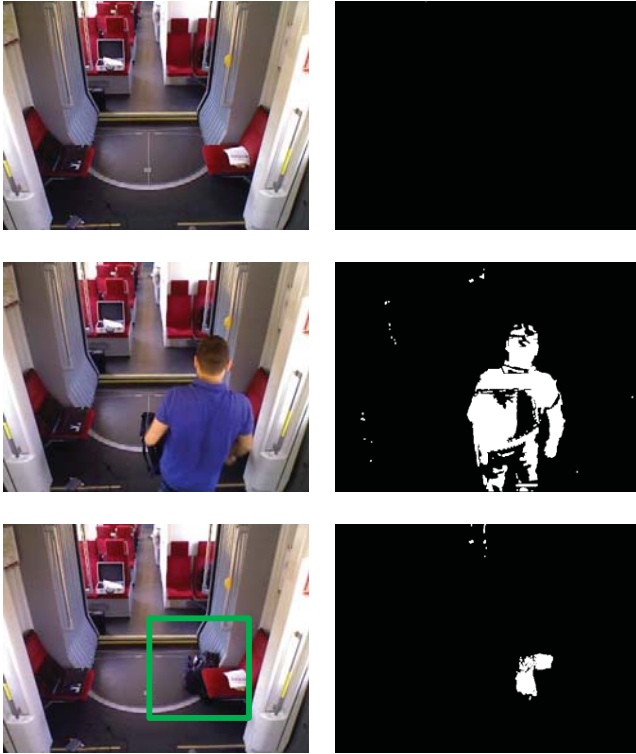


Figure 3: The live image and segmented image inside the tram during the system evaluation

Further more, the performance of the algorithm has been evaluated using a dataset consisting of eight OpenNI videos (ONI). These videos contain synchronised RGB and depth raw data from the Xtion Live Pro. The recordings were taken in two locations (see Figure 4) inside the tram and depicted 4 scenarios (S_i) for abandoning a piece of luggage (see Table 1). The dataset was gathered during the active time of the project where we were given access to the tram for our detection system. Due to the unique combination of purpose and format of these recordings (unattended luggage and ONI format), we were not able to compare our algorithm with other performances. We also see that our work could be a starting point of reference for future research.

Table 1: The generated tram dataset of ONI files.

	S_1	S_2	S_3	S_4
Suspected object displaced	no	no	yes	yes
Interference of camera view	no	yes	no	yes



(a) Location one

(b) Location two

Figure 4: The two locations where the RGB-D sensor has been installed

The threshold value of the dwelling time was set at 40 seconds during the evaluations, whereas the results show the real time that elapsed until a true-positive detection by the system (see Table 2). The evaluation tests had a false positive rate of 0%. This can be traced back to the limited amount of evaluation videos and the limitation of the number of actors inside a scene to two persons. The behaviour of the algorithm at a fully packed tram has not been tested during that phase of the implementation, since it has been established at the beginning of the project InREAKT that major critical situations occur in a relatively empty tram.

Table 2: Results of the evaluation of the system on the tram dataset in seconds

	S_1	S_2	S_3	S_4
Location one	45	52	120	140
Location two	43	48	106	120

5 Conclusion

The main challenge for the detection of the unattended luggage is isolating the luggage from the background and the dynamic objects, since background subtraction techniques are influenced by changes in dynamic scenery over time. Through the usage of two different sensor types, three background models for each sensor and a coordinated update mechanism for each background model, a very robust system was created. The usage of SURF features and the storage of the exact position of an object make it possible to track unattended luggage even in crowded situations. The suggested system is a reliable method for detecting unattended luggage over a long period of time and is able to cope with lighting changes, and crowded scenes, independent from the shape and colour of potentially dangerous objects. The usage of a system like this, could improve the reaction times of the security personnel, who can in turn initiate actions to save lives, which are threatened by potential explosive devices hidden inside these pieces of luggage.

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